ASSIGNMENT I - BASED ON SDG GOALS

Course Title: Machine Learning Course Code: ITA0609

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1. **DESIGN A CRIME PREDICTION SYSTEM USING MACHINE LEARNING**
   CRIME PREDICTION SYSTEM USING MACHINE LEARNING
   SDG 11 – SUSTAINABLE CITIES AND COMMUNITIES

1. PROBLEM UNDERSTANDING

Crime is one of the major challenges faced by modern cities. Crimes such as theft, burglary, robbery, assault, vehicle theft, and other criminal activities can affect the safety and quality of life of people. As the population of cities increases, law enforcement agencies face difficulties in monitoring large areas and analyzing a large amount of crime data manually.

Traditional crime prevention methods mainly depend on police investigation and manual analysis of previous crime records. However, identifying complex crime patterns from large datasets can be difficult and time-consuming. Crime occurrence may depend on various factors such as location, time, day of the week, previous crime history, population density, weather conditions, and area type.

Machine Learning provides an intelligent solution for analyzing historical crime data and identifying hidden patterns. A Crime Prediction System using Machine Learning can analyze previous crime records and predict the crime risk level of a particular location and time period. The system can classify areas into low-risk, medium-risk, and high-risk zones.

The main objective of this project is to design a Machine Learning-based Crime Prediction System that can analyze historical crime data and predict possible crime risk levels. The prediction results can support law enforcement agencies in planning police patrols, allocating resources, and taking preventive actions.

The system does not replace police officers or make decisions about individuals. Instead, it analyzes general crime patterns and provides data-driven information about areas and time periods where the crime risk may be higher.

This project contributes to SDG 11 – Sustainable Cities and Communities by supporting safer communities, smart city development, and improved public safety.

2. DATA COLLECTION AND PREPARATION

The performance of a Machine Learning model depends on the quality of the data used for training. Therefore, the first step in developing the Crime Prediction System is to collect reliable crime-related data.

Crime data can be collected from police departments, government crime databases, public safety organizations, open data portals, and publicly available datasets.

The important input parameters may include:

• Crime Type

• Location

• Date and Time

• Day of the Week

• Month and Year

• Population Density

• Weather Conditions

• Area Type

• Previous Crime Frequency

• Distance from Police Station

• Crime Risk Level

The crime type may include theft, burglary, robbery, assault, vehicle theft, and other criminal activities. Location is an important feature because different areas may have different crime patterns.

Time is also an important factor. Crime frequency may change depending on the time of the day. Similarly, crimes may occur more frequently during weekends, holidays, or particular months.

Data preprocessing is performed before applying Machine Learning algorithms.

Data Preparation Steps:

• Remove duplicate crime records.

• Handle missing values.

• Correct incorrect data entries.

• Convert date and time into useful features.

• Convert categorical values into numerical values.

• Normalize or standardize numerical features.

• Divide the dataset into training and testing data.

Date and time information can be converted into features such as hour, day, month, year, weekday, and weekend. This helps the Machine Learning model identify time-based crime patterns.

Categorical data such as crime type, area type, and weather conditions can be converted into numerical values using encoding techniques.

The dataset is generally divided into training and testing data. The training dataset is used to teach the model about historical crime patterns, while the testing dataset is used to evaluate the prediction performance of the model.

A properly cleaned and prepared dataset helps improve the accuracy and reliability of the Crime Prediction System.

3. ANALYSIS USING NUMPY

NumPy is a Python library used for numerical computation and data analysis. In the Crime Prediction System, NumPy can be used to perform statistical calculations and numerical operations on the crime dataset.

The dataset may contain thousands of crime records. NumPy helps process numerical values efficiently and supports mathematical operations required during Machine Learning preprocessing.

Operations Performed Using NumPy:

• Calculate the total number of crime cases.

• Calculate the average crime frequency.

• Find the maximum and minimum crime frequency.

• Calculate crime frequency for different locations.

• Analyze monthly and yearly crime trends.

• Calculate mean, variance, and standard deviation.

• Identify areas with high crime frequency.

• Perform numerical operations for data preprocessing.

The average crime frequency can be used to understand the general crime level in different areas. Maximum and minimum values can help identify areas with the highest and lowest crime occurrences.

NumPy can also be used to analyze crime patterns based on time. For example, the number of crimes occurring during morning, afternoon, evening, and night can be compared.

The results obtained from numerical analysis are used for further Machine Learning model development.

4. MACHINE LEARNING MODEL DEVELOPMENT

After data collection and preprocessing, the prepared dataset is given as input to Machine Learning algorithms. The algorithm learns patterns from historical crime data and predicts the crime risk level for new data.

The following Machine Learning algorithms can be used:

Random Forest:

Random Forest combines multiple Decision Trees to produce a final prediction. It can analyze several features such as location, time, previous crime frequency, and population density. It is suitable for classifying crime risk into low, medium, and high categories.

Decision Tree:

A Decision Tree makes predictions using a tree-like structure. It makes decisions based on different input features. For example, if a location has high previous crime frequency and the time is a peak period, the system may predict a high crime risk.

Logistic Regression:

Logistic Regression can be used for classification problems. It can predict whether the crime risk is low or high based on the input features.

Support Vector Machine:

Support Vector Machine can separate different crime risk categories by identifying a suitable decision boundary.

K-Nearest Neighbors:

KNN predicts the crime risk of a new location by comparing it with similar historical crime records.

Among these algorithms, Random Forest and Decision Tree can be useful because they can handle multiple features and provide understandable predictions.

5. CRIME PREDICTION PROCESS

The proposed Crime Prediction System follows a systematic Machine Learning process.

Step 1: Data Collection

Historical crime data is collected from reliable sources.

Step 2: Data Preprocessing

The collected data is cleaned, missing values are handled, and categorical data is converted into numerical form.

Step 3: Feature Extraction

Important information such as location, hour, month, day, and crime frequency is extracted.

Step 4: Numerical Analysis

NumPy is used to calculate statistical values and analyze crime patterns.

Step 5: Model Training

The training dataset is provided to the selected Machine Learning algorithm.

Step 6: Model Testing

The trained model is tested using unseen testing data.

Step 7: Crime Risk Prediction

The system predicts the crime risk level for a particular location and time.

Step 8: Visualization

The prediction results are represented using graphs, charts, and maps.

For example, the system may receive the following input:

Location: Area A

Time: 8:00 PM

Day: Saturday

Previous Crime Frequency: High

Population Density: High

The output of the system may be:

Predicted Crime Risk Level: HIGH

This information can support authorities in planning preventive measures and improving monitoring in the area.

6. VISUALIZATION AND INTERPRETATION

Data visualization helps understand crime patterns and trends more easily. Different graphs and charts can be used to represent the results of the analysis.

Graphs Used:

Line Graph – Crime Frequency vs Time

A line graph can show the variation in crime frequency over different time periods. It can help identify increasing or decreasing crime trends.

Bar Chart – Number of Crimes by Location

A bar chart can compare crime frequency in different locations and identify high-crime areas.

Pie Chart – Distribution of Crime Types

A pie chart can show the percentage of different crime types, such as theft, burglary, robbery, and vehicle theft.

Heat Map – Crime Density Across Areas

A heat map can identify crime hotspots. Areas with high crime density can be easily identified for additional monitoring.

Scatter Plot – Population Density vs Crime Frequency

A scatter plot can help analyze the relationship between population density and crime occurrence.

The visualizations help identify:

• High-risk locations.

• Peak crime hours.

• Common types of crimes.

• Monthly and yearly crime trends.

• Areas requiring increased police patrolling.

Visualization provides law enforcement agencies with useful information for better decision-making.

7. MODEL EVALUATION

The performance of the Crime Prediction System must be evaluated after training the Machine Learning model.

The following evaluation metrics can be used:

Accuracy:

Accuracy measures the percentage of correct predictions made by the model.

Precision:

Precision measures how many predicted high-risk cases are actually correct.

Recall:

Recall measures how many actual high-risk cases are correctly identified.

F1-Score:

F1-score provides a balanced measurement of precision and recall.

Confusion Matrix:

A confusion matrix shows the number of correct and incorrect predictions for different crime risk categories.

The Machine Learning model with the best performance can be selected for the final Crime Prediction System.

8. INNOVATION AND SDG MAPPING

Innovation:

The proposed system introduces an intelligent data-driven method for crime pattern analysis and risk prediction.

The system can:

• Predict crime risk levels in different areas.

• Identify crime-prone locations.

• Analyze crime patterns based on time and location.

• Support police patrol planning.

• Improve allocation of law enforcement resources.

• Provide crime trend visualization.

• Support smart city security systems.

• Provide data-driven information for preventive action.

SDG 11 – Sustainable Cities and Communities:

This project directly contributes to SDG 11, which focuses on making cities and human settlements inclusive, safe, resilient, and sustainable.

Contribution to SDG 11:

• Improves public safety.

• Helps identify crime-prone areas.

• Supports intelligent policing.

• Improves resource allocation.

• Supports smart city development.

• Helps create safer communities.

• Promotes data-driven urban management.

By using Machine Learning for crime analysis, cities can develop intelligent systems for improving public safety and urban security.

9. APPLICATIONS AND ADVANTAGES

Applications:

• Smart City Security Systems.

• Police Patrol Planning.

• Crime Hotspot Identification.

• Public Safety Monitoring.

• Urban Planning.

• Emergency Response Planning.

• Crime Trend Analysis.

Advantages:

• Analyzes large amounts of historical crime data.

• Helps identify high-risk locations.

• Supports preventive policing.

• Reduces manual data analysis.

• Improves resource allocation.

• Provides visual crime pattern analysis.

• Supports smart city development.

• Helps identify crime trends.

10. DOCUMENTATION AND CONCLUSION

The complete project documentation includes the problem statement, objectives, dataset description, data collection, preprocessing methods, NumPy-based analysis, Machine Learning model development, visualization, model evaluation, innovation, applications, and SDG mapping.

The Crime Prediction System uses historical crime data to identify patterns and predict crime risk levels. Data is first collected and cleaned. Important features such as location, time, crime type, and previous crime frequency are extracted. NumPy is used for numerical analysis and statistical calculations.

Machine Learning algorithms such as Decision Tree, Random Forest, Logistic Regression, SVM, and KNN can be used to develop the prediction model. The trained model predicts the crime risk level for a particular location and time period.

Data visualization techniques such as line graphs, bar charts, pie charts, scatter plots, and heat maps help understand crime trends and identify high-risk areas.

Crime Prediction System Using Machine Learning is an effective approach for improving public safety and supporting intelligent urban management. The system helps law enforcement agencies analyze crime patterns and make better data-driven decisions.

The project significantly contributes to SDG 11 – Sustainable Cities and Communities by supporting safer cities, intelligent policing, and sustainable urban development. By combining Machine Learning, NumPy-based numerical analysis, and data visualization, the proposed system provides a useful solution for analyzing crime patterns and supporting preventive safety measures.

In the future, the system can be improved by integrating real-time crime data, Geographic Information Systems (GIS), IoT-based monitoring systems, and advanced Machine Learning algorithms. With responsible and ethical use, Machine Learning-based crime prediction can support the development of safer, smarter, and more sustainable communities.